DASHBOARD OPERATING GUIDE

NYC311 Workforce Control

Supervisor-facing daily plan, safe capacity chart, release checks, and JSON export

Prepared 2026-10-24 | Detailed user, data, model, and operational reference

Local URL	http://localhost:8052/
Launch	.venv/bin/python src/workforce_control_app.py
Source	Current public service-request aggregates; not call-center ACD data
Plan export	JSON download named nyc311-staffing-<plan-date>.json

Important data and decision boundary

NYC 311 service-request records are not equivalent to phone-call arrivals, agent schedules, or payroll. Forecast conversions and labor inputs are scenarios unless calibrated against authorized operational records. Dashboard output is planning support and requires human review.

1. Purpose and page structure

Workforce Control is a separate supervisor-oriented daily planning dashboard inspired by the supplied site extraction. It presents source freshness, plan date and status, a conditional risk alert, forecast and staffing KPIs, an hourly coverage chart, operating controls, a recommended shift table, data/safety checks, and a downloadable JSON scenario.

The interface uses the reference's dark operations palette and compact status cards. It does not reproduce the reference's September 2026 simulated snapshot or its browser-side greedy calculation. It uses the current NYC Open Data hourly aggregate and the repository's server-side mixed-integer optimizer.

All exported staffing labels and metrics are planning scenarios. The current public source measures service requests, not call arrivals or agent work. The page labels this distinction and asks supervisors to calibrate the request-to-call factor from operational data.

2. Start-up, source, and forecast

Start the app from the repository root. It loads 120 days of hourly NYC 311 service-request counts from the public Socrata endpoint, grouped by date, hour, and borough. The header shows the latest source date. Refresh live request data to replace the browser-held data; refresh failures are shown rather than ignored.

The forecast aggregates the most recent 56 days, matches historical weekdays to the selected plan date, and calculates the 75th percentile independently for each hour. It uses at least four matching weekdays. The resulting

request proxy is scaled by the editable calls-per-request scenario factor; the default is 0.05.

A 28-day rolling-origin validation window reports MAE and RMSE in scenario calls per hour. These values describe the historical backtest of the same weekday/hour percentile method; they are not a production service-level guarantee.

- Default plan date is at least tomorrow in America/New_York and after the latest source date.
- The borough menu includes ALL and the borough categories available in the NYC 311 source.
- The 75th percentile is a conservative scenario input, not a calibrated probability interval.

3. Scenario inputs

Daily labor budget ranges from $5,000 to $25,000 in $250 increments. The hourly per-agent capacity control ranges from 4 to 12. Demand buffer ranges from 100% to 130%. Per-agent shift maximum ranges from 32 to 72 calls in steps of two; the optional minimum is disabled at zero and ranges up to 32.

Additional inputs are plan date, borough, calls-per-request factor, available agent count, and hourly wage. Shifts are modeled as eight hours and may start at 00:00, 04:00, 08:00, 12:00, or 16:00. A midpoint hour is modeled at half capacity to represent a simplified break.

The initial scenario uses a $15,500 budget, eight calls per agent-hour, a 112% buffer, a 52-call per-shift maximum, no minimum workload, 0.05 calls per service request, 35 interchangeable agents, and $31.50 per paid hour. Every assumption is visible in the JSON export. The available-agent pool and hourly wage are not supplied by the public 311 dataset; populate those values from authorized workforce and payroll records before evaluating an operational proposal.

4. Optimization, status, and visual outputs

The scenario applies the demand buffer, then passes the buffered hourly vector to the existing SciPy HiGHS MILP. The model minimizes uncovered hourly demand first and complete-shift labor cost second, subject to the hard budget, hourly capacity, per-shift workload limits, one shift per agent, and available staffing.

The chart compares the unbuffered call scenario and buffered demand to effective scheduled capacity after modeled breaks, and marks uncovered buffered calls. The peak note reports the peak forecast service-request proxy and rolling validation metrics.

Summary cards show forecast service-request volume, selected shifts, modeled spend, remaining budget, uncovered demand, and coverage percentage. Shift rows group selected agents by start time and show agent count, assigned calls, average calls per agent, and a Balanced, Near cap, or Light badge.

Ready for review is displayed only when no demand remains uncovered and the solver reports optimality. Otherwise the dashboard marks the plan Action required and surfaces the modeled shortfall or solver state. A feasible result is not an approval or a guarantee of real-world coverage.

5. Review and export workflow

Confirm the source date and borough; adjust the demand conversion, buffer, availability, wage, and safe capacity assumptions; then select Calculate scenario. Review validation error, peak hour, coverage, spend, remaining budget, shift loads, and solver status before downloading the JSON plan.

The export contains generated time, source description and date, plan date, borough, all scenario inputs, summary metrics, hourly forecast/capacity/shortfall rows, shift totals, and validation metrics. It is a downloadable scenario artifact only. The dashboard does not persist, approve, audit, authenticate, or publish schedules.

If uncovered demand remains, consider an authorized budget increase, verified additional availability, or an approved overflow path. Do not exceed safe workloads, contractual limits, or break rules to eliminate a warning.

6. Limitations and production readiness

Service requests and phone calls are distinct event types. The model has no live queue balance, abandonment, service-level target, agent skills, actual breaks, contractual rest or overtime rules, cross-midnight shifts, or locked-in assignments. The single-day shortfall metric means modeled demand not allocated in the same hour; it is not a measure of calls lost.

Before operational use, replace the proxy with validated ACD arrivals, estimate handling and occupancy from telemetry, construct truly feasible individual shifts, represent breaks and labor rules, include forecast uncertainty and queue dynamics, and obtain supervisor approval. Add access control, audit logging, schedule persistence, monitoring, and recovery before any production release.

The local Dash development server is for evaluation only. Use a production WSGI deployment and appropriate access controls if this prototype is hosted beyond a trusted local environment.